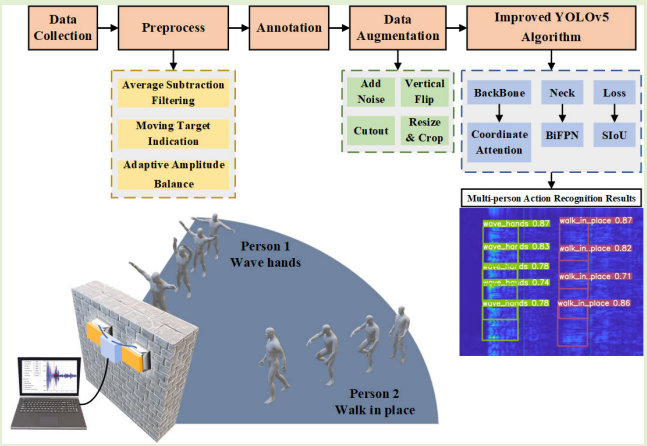
Through-Wall Multi-Person Action Recognition Using Enhanced YOLOv5 and IR-UWB Radar

Xiaoxiong Li, Shuning Zhang, Si Chen, *Senior Member, IEEE*, and Zelong Xiao

***Abstract*—Impulse radio ultrawideband (IR-UWB) radar has been extensively utilized in recognizing human activities due to its high resolution, independence from light and occlusion, and nonintrusive nature. However, current research is mainly limited to single-person and line-of-sight conditions, and many challenges exist in multi-person action recognition under non-line-of-sight (NLOS) conditions. This article proposes a through-wall, multi-person action recognition framework based on the improved YOLOv5 network. First, we collected the through-wall multi-person motion echoes using the IR-UWB radar. After processing them with algorithms such as clutter suppression and adaptive amplitude balancing, we successfully constructed a dataset containing time–range images of multi-person actions. Then, to improve the detection and classification performance, we improved the YOLOv5 network and introduced the bi-directional feature pyramid network (BiFPN) module, SCYLLA-intersection over union (SIoU) loss function, and coordinate attention (CA) mechanism. The BiFPN module enhances the model's learning ability for multiscale features. The SIoU loss function optimizes the positioning accuracy of the prediction box. The CA attention mechanism finely weights feature coordinates, enhancing the feature discrimination ability. The experimental results show that the improved YOLOv5 network achieves good detection and classification results on our dataset. Moreover, the proposed method has verified good generalization ability on the moving and stationary target acquisition and recognition (MSTAR) public dataset.**



***Index Terms*— Attention mechanism, human activity recognition (HAR), multiple targets, through-wall radar, ultrawideband (UWB) radar.**

I. Introduction

HUMAN activity recognition (HAR) has important applications in many fields, including health monitoring, security surveillance, smart homes, and more. Over the past few decades, methods based on optical sensors, such as cameras, have greatly progressed in HAR [1]. Representative visual action recognition algorithms include recurrent neural networks (RNNs) and convolutional neural networks (CNNs) [2], [3]. These methods can exploit the spatial–temporal information of images and videos for action representation and understanding. In addition, accelerometers and gyroscopes in wearable devices are widely used for human motion capture and recognition [4]. However, these studies were often conducted in controlled environments with ideal lighting and unobstructed views. In real-world scenarios, challenges, such as non-line-of-sight (NLOS) conditions, insufficient lighting, or the inconvenience of wearing equipment, can significantly affect the system performance. Recent studies have begun to address HAR in more complex environments. For example, Fontaine et al. [5] investigated error correction and NLOS classification in multiple environments using transfer learning techniques. Che et al. [6] proposed a fine-tuned attribute-weighted Naïve Bayes classifier for positioning in NLOS conditions. Kim and Pyun [7] developed an enhanced ranging method utilizing a denoising neural network to improve accuracy under noisy conditions.

Impulse radio ultrawideband (IR-UWB) radar has been extensively utilized in automatic target recognition due to its high resolution and strong penetration capability [8], [9],

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[10], [11]. Due to its insensitivity to environmental changes, IR-UWB radar has been gradually applied in HAR. Compared with visual cameras, it effectively overcomes the challenges caused by obstructions and lighting changes. Compared with wearable devices, it does not require tags and is more suitable for noninvasive monitoring. IR-UWB radar emits extremely short pulse signals. By analyzing the spatial static and temporal dynamic information of the echo sequence, high-precision motion capture can be achieved. Lai et al. [12] utilized a method where segments of time–Doppler images were fed sequentially into a long short-term memory network (LSTM) to classify sleep postures. Yang et al. [13] developed a CNN incorporating a dynamic weight attention mechanism using time–Doppler spectrum images of particular movements to identify individuals. Beaulieu et al. [14] merged the feature vectors obtained from EfficientNet and LSTM models and then used the fully connected network to identify human activities. Despite significant advancements in HAR using IR-UWB radar, current research mainly focuses on single people and line-of-sight conditions. Multi-Person action recognition under NLOS conditions has not been fully studied. multi-person movement monitoring in complex environments is very necessary for practical applications. For example, rescue robots must simultaneously locate multiple survivors while exploring the ruins. Similarly, for urban security and street fighting, monitoring the intrusion actions of multiple targets through walls is essential.

To solve the problem of multi-person action recognition under through-wall conditions, it is necessary to detect the target position in the radar feature map and extract the action discrimination features of the target area. You only look once (YOLO) is a mainstream single-stage target detection network that considers speed and accuracy [15]. In contrast to traditional two-stage detection methods, YOLOs simultaneously predict the location and category of the target, avoiding the accumulation of errors between area detection and classification. This feature of YOLO makes it very suitable for handling multiobject detection and classification in time–range images. Its iterative version, YOLOv5, adopts the backbone–neck–head network structure design. The backbone consists of CSPDarknet53 and is used for feature extraction. The neck incorporates path aggregation network (PANet) and spatial pyramid pooling (SPP) modules to fuse feature information at different scales. The head is tasked with predicting the position and classification results of the box. However, YOLOv5 is mainly designed for optical images. Its original structure may have limitations for the special needs of IR-UWB radar data and multi-person action recognition through walls. For example, when YOLOv5 processes radar time–range images, it may have insufficient ability to integrate multiscale features or need to be improved regarding localization accuracy and the discriminability of feature expression.

Based on the above analysis, this article proposes a through-wall multi-person activity recognition approach based on IR-UWB radar and the YOLOv5 network. First, we use IR-UWB radar to collect echo signals of multiple people's actions under wall penetration conditions. Then, the echo signal is preprocessed, including removing dc components, background elimination, amplitude balancing, to form time–range images input to the network. Next, to simultaneously locate the time–range area where each person is located and classify their action categories, we introduce an enhanced YOLOv5 network. It includes the following three improvements: 1) SCYLLA-intersection over union (SIoU) loss function to improve the positioning accuracy of the prediction frame; 2) bi-directional feature pyramid network (BiFPN) module to enhance the model's integration of multiscale features; and 3) coordinate attention (CA) mechanism to improve the discriminability of temporal distance feature expression. Finally, we verified the effectiveness and generalization performance of the method on the self-built and the moving and stationary target acquisition and recognition (MSTAR) public datasets. Experimental findings reveal that the improved YOLOv5 model can achieve robust multi-person action detection and recognition under through-wall conditions and has good generalization performance.

The article's main contributions are as follows.

1) A pioneering exploration of the IR-UWB radar-based joint recognition problem for through-wall multi-person actions and the construction of the corresponding dataset, which has important application value.
2) The adaptive amplitude balance algorithm alleviates the amplitude imbalance problem caused by distance, action type, and other factors and better highlights the characteristics of the target area.
3) Based on the YOLOv5 network, we introduce the BiFPN module, SIoU loss function, and CA attention mechanism to form a detection and classification network suitable for processing radar time–range images. It elevates the model's proficiency in understanding time–range images and adaptability in complex environments.
4) The effectiveness of the proposed algorithm was verified on self-built datasets and public datasets.

The rest of this article is structured as follows. Section II provides the related work. Section III elaborates on the proposed framework and method in detail. Section IV gives the experimental results and analysis. Section V discusses the limitations and future work. Finally, Section VI succinctly concludes this work.

II. Related Work

A. Principle of IR-UWB Radar

Usually, the transmission signal bandwidth of ultrawideband (UWB) radar will exceed 500 MHz or its relative bandwidth will be greater than 20%. IR-UWB radar emits broadband pulse signals and obtains the target's spatial static and temporal dynamic information based on the received echo sequence. Since distance resolution is proportional to signal bandwidth, higher bandwidth allows for finer measurements. Compared to radars of other regimes, IR-UWB radar features an extremely narrow pulsewidth and has the characteristics of high resolution, strong resistance to multipath effects, and strong through-wall detection capabilities [16]. This makes it ideal for human activity monitoring and recognition.

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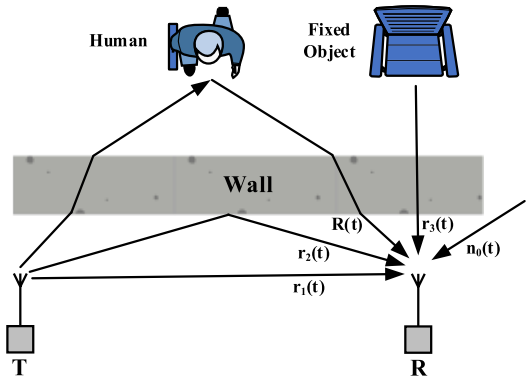


Fig. 1. IR-UWB radar wall-penetrating echo model.

As shown in Fig. 1, the through-wall echo model of IR-UWB radar shows multiple components of the received signal. These components include the target reflection as well as several other components and can be expressed by the following expression [17]:

$$E(t) = R(t) + r_1(t) + r_2(t) + r_3(t) + n_0(t). \tag{1}$$

Among them, $R(t)$ is the reflected wave from the human target, which we are mainly concerned about. $r_1(t)$ is the direct wave coupled by the antenna, $r_2(t)$ is the reflected wave from the wall, and $r_3(t)$ is the reflected wave from the fixed object. $n_0(t)$ is the noise produced by various interferences in the environment.

For IR-UWB radar, each frame (mth frame) echo signal of the human target is the synthesis of echoes from its multiple scattering centers [18]

$$R_m^{\text{frame}}(t) = \sum_{i=1}^{I} a_i P_m(t - \tau_i) \tag{2}$$

where $P(t)$ is the transmitted pulse signal; I is the number of scattering centers; and a_i and τ_i are the amplitude attenuation and time shift of the ith scattering point, respectively. The continuous multiframe radar echo signals can be represented as a matrix

$$\boldsymbol{R}_{m,n} = R_m^{\text{frame}}(nT_f), \quad m = 1, 2, \ldots, M; \ n = 1, 2, \ldots, N \tag{3}$$

where T_f represents the interval for sampling each frame. The dimension m is often called the "slow-time" dimension, with the gap between successive frames being the pulse repetition interval. n is the "fast-time" dimension, which is the discrete sampling points within a frame of signal. Regarding human motion, the fast-time dimension seizes the spatial aspects of body posture, whereas the slow-time dimension chronicles the time-related traits of human movement. Similar to camera video or skeleton movements, an echo matrix composed of multiple frame echoes can similarly depict the spatiotemporal features of human body movements. After preprocessing and other operations, the above echo matrix $\boldsymbol{R}_{m,n}$ can be converted into a time–distance image, which can be used as input information for action recognition. The time–distance image reflects the movement posture and trajectory of the target in a 2-D form and represents human body movements more intuitively. This article uses time–distance images to detect and classify multi-person actions.

B. IR-UWB Radar Single-Person Action Recognition

HAR via IR-UWB radar involves three key phases: 1) preprocessing of the signal; 2) extraction of significant features; and 3) identification through classifiers. In the preprocessing phase, techniques, such as average subtraction filtering and moving target indicator (MTI), are commonly employed [19]. The extracted features from radar echoes typically focus on aspects such as time-domain attributes [20], Doppler characteristics [13], time–range mapping [21], [22], and range–Doppler map [23], among others. Regarding classifiers, there are a range of methods from traditional ones such as KNNs [24], SVMs [25], and decision trees [26] to more advanced options such as CNNs [27], [28], [29] and RNNs [30], [31], [32]. In these works, Liu et al. [33] analyzed four distinct features from the time–range matrix of six types of human interactive movements, finding that the KNN approach outperformed several deep learning models in accuracy, particularly when dealing with limited data samples. Another study [34] introduced a segmented convolutional gated RNN model, utilizing the Doppler signature image as its input for recognizing human activities. Furthermore, Ma et al. [35] merged the strengths of CNNs with convolutional LSTM networks to derive spatial and temporal characteristics from radar echoes for applications such as detecting falls. Finally, Huang and colleagues developed an autoencoder network to detect humans in challenging weather conditions and areas covered with foliage [36]. IR-UWB radar action recognition research currently uses deep learning algorithms to classify feature images (such as time–distance, range–Doppler, and time–Doppler) of various single-person actions. These methods are only suitable for single-person simple action classification and cannot be extended to multi-person action recognition in complex environments.

C. YOLOv5 Network

Compared with traditional target detection techniques of region extraction and classification, the YOLO algorithm can simultaneously predict the category and location of the target in a single stage with high speed and accuracy. YOLOv5 is its optimized iterative version, proposed by Ultralytics Laboratory in 2020 [37], and performs well on COCO, VOC, and various actual detection tasks. Its backbone usually contains a cross-stage partial (CSP) structure. CSP introduces cross-layer channel segmentation and path aggregation, which can alleviate the vanishing gradient problem and maintain strong feature expression capabilities. The neck part includes feature pyramid network (FPN), PANet, and SPP structures, which can fuse semantic information at different scales. The head part predicts the category and location of the target. The loss function of YOLOv5 includes three parts: confidence binary cross-entropy (BCE) loss, classification BCE loss, and positioning complete

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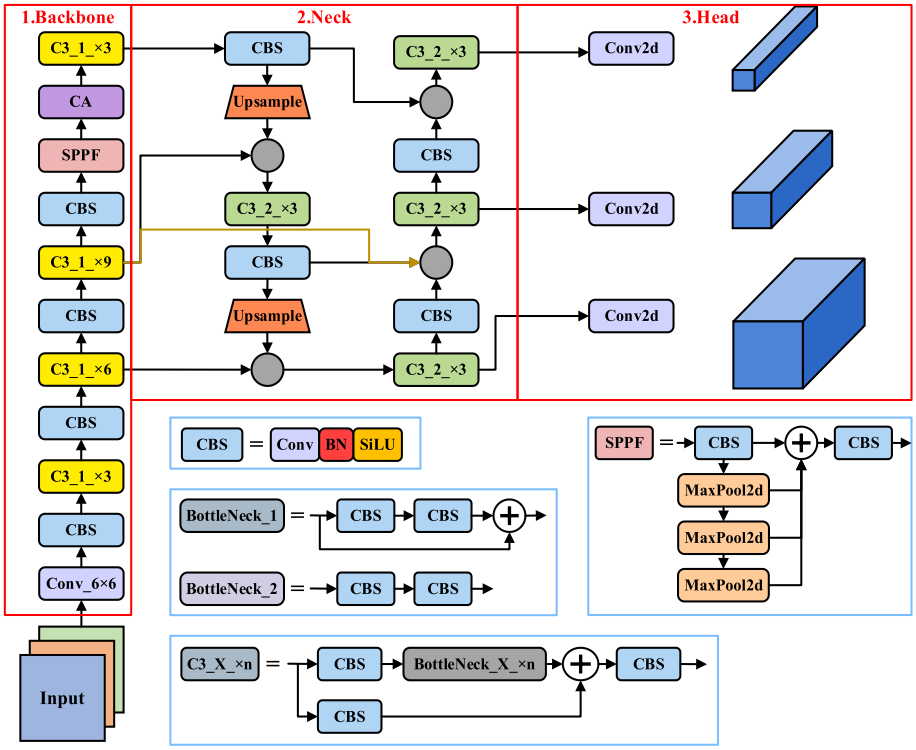


Fig. 2. Improved YOLOv5 network structure diagram.

IoU (CIoU) loss. The YOLOv5 algorithm also introduces various technologies for performance optimization, such as mosaic data enhancement, multiscale training, and adaptive anchor frame calculation.

Compared with classification networks, YOLOv5, as a detection network, is more suitable for processing radar feature images containing multiple targets. It can simultaneously locate each target and give classification results, fulfilling the multi-person action recognition task requirements discussed in this article. Therefore, we choose YOLOv5 as the basic framework and make key improvements based on the characteristics of radar data, which will be described in Section III.

III. Method

A. Overall Architecture

The through-wall multi-person action recognition network proposed in this article has been developed by enhancing the YOLOv5 framework, as illustrated in Fig. 2. The network can be divided into three stages: backbone feature extraction part, neck feature fusion part, and prediction part. The network input is the preprocessed time–distance feature map of multi-person activities.

Backbone follows the CSP structure and introduces the CA attention mechanism [38]. In our implementation, we have adopted a simplified version of CSP, called C3, which omits one convolution operation to streamline the architecture and accelerate inference. The CA attention module was added before the last C3 layer to enhance the backbone's feature expression ability. It captures long-range dependencies in one spatial direction while retaining precise position information in another spatial direction. By encoding a pair of direction-aware and position-sensitive attention maps and applying them complementarily to the input feature map, the CA attention module can effectively enhance the representation of the object of interest, thereby improving the network's ability to recognize complex actions.

In the neck part, we also achieved innovation and introduced the BiFPN multiscale feature fusion module [39]. BiFPN aggregates the multiscale feature information of backbone through the design of bidirectional feature flow, further enhancing the network's representation ability. This design enables the model to flexibly capture the spatiotemporal features at different scales when processing human motion feature images, thereby effectively improving the overall robustness of the model.

The prediction part contains action category, confidence, and location box prediction. To ensure prediction accuracy, we simultaneously train the network using confidence loss, classification loss, and SIoU loss [40]. The SIoU loss function optimizes bounding box regression by introducing geometric information such as angle and distance, aiming to improve the accuracy and speed of the predicted box approaching the true box.

Integrating CA, BiFPN, and SIoU loss improves the network's processing of complex time–range images. The principles and functions of each component are explained in detail next.

B. CA Attention Mechanism

Due to distance attenuation or occlusion and interference caused by complex environments, the time–range image of IR-UWB radar has high noise, which brings difficulties to

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Algorithm 1 Adaptive Amplitude Balance

Input: Echo matrix $X_{m,n}, m = 1, 2, \ldots, M; n = 1, 2, \ldots, N$, segment length L, maximum gain $G_{\max}$, overlap rate γ.
Output: The processed matrix $\hat{X}_{m,n}$.
1: Calculate slow time points (M), distance points (N), overlap length ($L_{overlap} = \lceil L \cdot \gamma \rceil$), and number of segments ($N_{seg} = \left\lceil \frac{N - L_{overlap}}{L - L_{overlap}} \right\rceil$).
2: Fill $X_{m,n}$ with all-zero columns (still noted as $X_{m,n}$)
3: For each slow time point i, do:
  a: Initialize $\hat{X}_{i,n}$ and $G_0^i(j), 1 \le j \le N_{seg}$
  b: For each segment j, do:
    i: Calculate start and end indices ($start = (j-1) \cdot (L - L_{overlap}) + 1$; $end = start + L - 1$)
    ii: Cut segment $X_{i,n}, start \le n \le end$, calculate $G_0^i(j) = L / \sqrt{\sum_{n=start}^{end} X_{i,n}^2}$
  c: Normalize $G_0^i(j)$: $G_{norm}^i(j) = \frac{G_0^i(j)}{min(G_0^i(j))}$, set values in $G_{norm}^i(j)$ greater than $G_{\max}$ to $G_{\max}$.
  d: For each segment j, do:
    i: Calculate start and end indices (same as b→i)
    ii: Cut segment $X_{i,n}, start \le n \le end$, multiply by corresponding $G_{norm}^i(j)$, and store in $Y_{i,n}, start \le n \le end$.
4: Return $\hat{X}_{m,n} = Y_{m,n}, n = 1, 2, \ldots, N$.

action representation learning. The attention mechanism can enhance the model's attention to important parts of the input and suppress the interference of irrelevant information by learning the global dependencies of input features. The CA module efficiently gathers information across channels while being sensitive to direction and position. It enhances the model's precision in pinpointing and recognizing the human subject being tested. Integrating the CA module just before the final C3 module of the backbone boosts its proficiency in interpreting intricate time–distance feature images. Fig. 3 illustrates the CA module's design. Initially, two unidimensional global pooling processes collect the incoming features from vertical and horizontal perspectives, forming two distinct direction-aware feature maps. Following this, the two direction-specific feature maps are transformed into a pair of attention maps. Each of these maps is designed to understand the extensive dependencies of the input features along a specific spatial axis, effectively retaining locational data within the resultant attention map. Ultimately, by multiplying these attention weights with the original feature map, the overall representative capability of the feature map is significantly enhanced.

Specifically, two pooling kernels, $(H, 1)$ and $(1, W)$, are applied to the input features $\boldsymbol{x} \in \mathbb{R}^{C \times H \times W}$, encoding each channel along the X-axis and Y-axis, respectively. Consequently, the output for the cth channel at a specific height h and width w can be described as follows:

$$\boldsymbol{z}^h = z_c^h(h) = \frac{1}{W} \sum_{0 \le i < W} x_c(h, i), \quad c = 1, 2, \ldots, C \tag{4}$$

$$\boldsymbol{z}^w = z_c^w(w) = \frac{1}{H} \sum_{0 \le j < H} x_c(j, w), \quad c = 1, 2, \ldots, C. \tag{5}$$

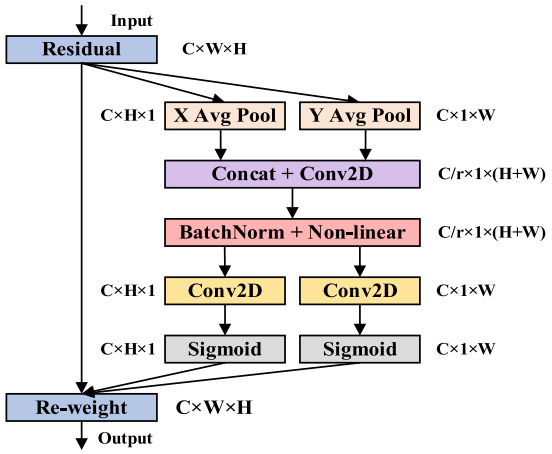


Fig. 3. Schematic of CA module structure.

Then, the above encoding feature maps are connected and passed through the convolution transformation with a kernel of 1×1 $F_1(\cdot)$ and the nonlinear activation function $\delta(\cdot)$, yielding

$$\mathbf{f} = \delta\left(F_1\left(\left[\mathbf{z}^h, \mathbf{z}^w\right]\right)\right). \tag{6}$$

$\mathbf{f} \in \mathbb{R}^{C/r \times (H+W)}$ is the intermediate feature map, in which r is the channel reduction ratio. Next, we divide $\mathbf{f}$ into $\mathbf{f}^h \in \mathbb{R}^{C/r \times H}$ and $\mathbf{f}^w \in \mathbb{R}^{C/r \times W}$ along the spatial dimension and pass through 1×1 convolution transformation functions $F_h(\cdot)$ and $F_w(\cdot)$, respectively. After passing the sigmoid function, we get

$$\mathbf{g}^h = \sigma\left(F_h\left(\mathbf{f}^h\right)\right) \tag{7}$$

$$\mathbf{g}^w = \sigma\left(F_w\left(\mathbf{f}^w\right)\right). \tag{8}$$

Finally, the outputs $\mathbf{g}^h$ and $\mathbf{g}^w$ are expanded and then applied as attention weights multiplied with the input $\boldsymbol{x}$. The output $\boldsymbol{y} \in \mathbb{R}^{C \times H \times W}$ of the CA block can be expressed as

$$y_c(i, j) = x_c(i, j) \times g_c^h(i) \times g_c^w(j) + x_c(i, j). \tag{9}$$

C. BiFPN Module

FPN achieves the effect of merging multiscale features through top–down sampling and bottom–up sampling paths. However, the flow of information is one way, and low-level features cannot be fed back to high-level features. In time–distance images, the actions performed by multiple individuals may exhibit significant variations in duration and spatial extent. To increase multiscale feature interaction, we use the BiFPN module in the neck part of YOLOv5, as shown in Fig. 4. BiFPN contains both a top–down path (TD) and a bottom–up path (BU). The downward path passes high-level semantic features to the lower layers through 1×1 convolution and upsampling, allowing the higher layers to guide the lower layers. The upward path employs 1×1 convolution and downsampling to transmit lower level, position-sensitive information upward, facilitating the refinement of higher layers by low-level features. In this way, high- and low-level features can achieve two-way interaction and fusion, producing richer and more stable multiscale features, which is very beneficial for

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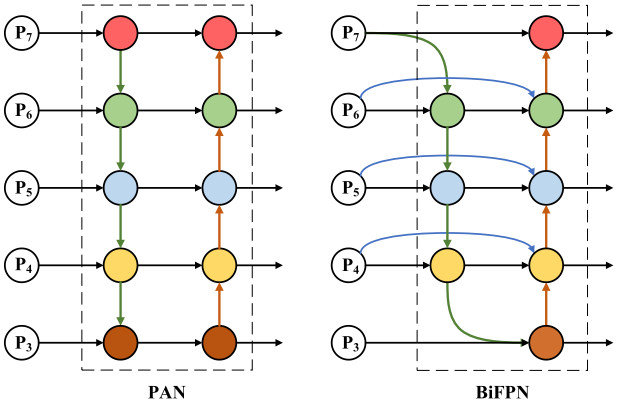


Fig. 4. Schematic of BiFPN structure.

extracting key information from multiple people in complex scenarios.

D. SIoU Loss Function

In target detection, the IoU ratio of the predicted and true boxes is an important indicator for evaluating the detection accuracy. Traditional object localization losses (such as generalized IoU (GIoU), distance-IoU (DIoU) [41], and CIoU [37]) are a combination of several bounding box regression metrics, such as the intersecting region between the forecast and true boxes, center point distance, and aspect ratio. The original YOLOv5 network uses CIoU as the positioning loss, considering the IoU, distance between center points, and aspect ratio of the predicted box and the true box.

However, none of these methods consider the mismatch in orientation between the ground truth box and the predicted box. Prediction boxes may "wander" during training, leading to slower convergence and poor model performance. Gevorgyan proposed a new loss function SIoU that considers the angle between the desired regression vectors. The SIoU loss function is defined as

$$\text{Loss}_{\text{SIoU}} = 1 - \text{IoU} + \frac{\Delta + \Omega}{2} \tag{10}$$

where IoU is the intersection-over-union ratio, Δ is the distance loss, and Ω is the shape loss. The following are introduced.

1) *IoU:* The ratio of the intersection over union between the predicted and true boxes.
2) *Distance Loss* Δ

$$\Delta = \sum_{t=x,y} \left(1 - \mathrm{e}^{-\gamma\rho_t}\right) = 2 - \mathrm{e}^{-\gamma\rho_x} - \mathrm{e}^{-\gamma\rho_y} \tag{11}$$

where $\rho_x = ((b_{c_x}^{gt} - b_{c_x})/c_w)^2$ and $\rho_y = ((b_{c_y}^{gt} - b_{c_y})/c_h)^2$, in which c_w and c_h are the width and height differences between the center points of the true box and the predicted box, respectively, and (b_{c_x}, b_{c_y}) and $(b_{c_x}^{gt}, b_{c_y}^{gt})$ are the coordinates of the predicted and true boxes, respectively. $\gamma = 2 - \Lambda$, in which Λ is the angle loss

$$\Lambda = 1 - 2 * \sin^2\left(\arcsin\left(\frac{c_h}{\sigma}\right) - \frac{\pi}{4}\right) \tag{12}$$

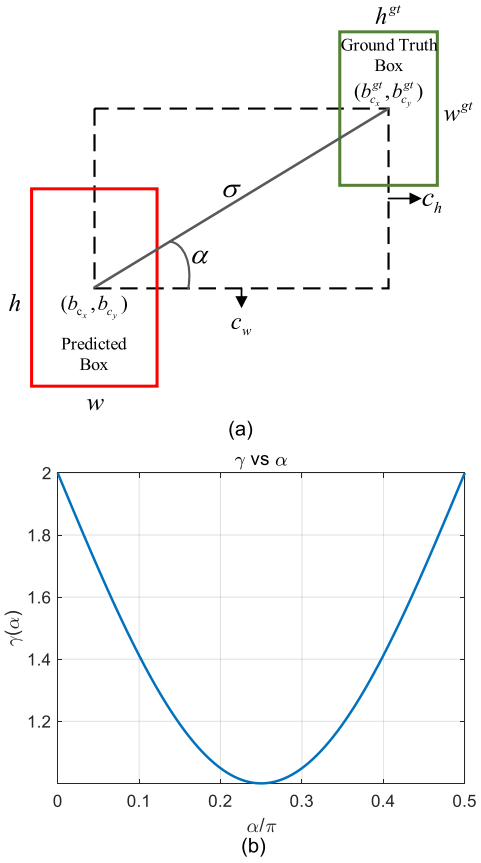


Fig. 5. SIoU loss. (a) Definition of relevant parameters. (b) Relationship between α and γ.

where σ is the distance between the center points of the true box and the predicted box. The relevant variables are shown in Fig. 5(a), and the relationship between α and γ is shown in Fig. 5(b). It can be seen that when $\alpha \to 0$ or $(\pi/2)$, $\gamma \to 2$, the contribution of distance cost decreases. When $\alpha \to (\pi/4)$, $\gamma \to 1$, the contribution of distance loss increases. Therefore, as the angle increases, the priority of γ is higher than the distance value. During the training process, the predicted box will first drift to the nearest axis and then get closer to the true box faster with fewer degrees of freedom.

The shape loss is

$$\Omega = \sum_{t=w,h} \left(1 - e^{-wt}\right)^{\theta} = \left(1 - e^{-w_w}\right)^{\theta} + \left(1 - e^{-w_h}\right)^{\theta}. \tag{13}$$

Among them, $w_w = (|w - w^{gt}|/(\max(w, w^{gt})))$ and $w_h = (|h - h^{gt}|/(\max(h, h^{gt})))$. (w, h) and (w^{gt}, h^{gt}) are the predicted and true boxes' width and height, respectively. Parameter θ controls how much attention is paid to shape loss. We substitute the CIoU loss with SIoU box regression loss in the original YOLOv5 framework to improve the accuracy and speed of the prediction box close to the true value box.

IV. Experiments and Results

Most current public IR-UWB radar human action datasets are single-person action data under line-of-sight conditions, which cannot satisfy algorithm research in complex scenes.

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TABLE I
NAMES AND DESCRIPTIONS OF ALL ACTIONS

Num	Action Name	Description
1	Bend and stand	The transition between bend and stand
2	Boxing	Punching movements
3	Run forward	Running forward
4	Run in place	Running in place
5	Sit and stand	The transition between sit and stand
6	Walk forward	Walking forward
7	Walk in place	Walking in place
8	Wave hands	Waving hands for help

To validate the effectiveness of the proposed method, we collected multi-person action echo signals captured by IR-UWB radar under wall-penetrating conditions and constructed a dataset consisting of time–range images after preprocessing and data annotation. After performing data enhancements such as adding Gaussian noise, random cropping, and vertical flipping, we compared the performance of the improved YOLOv5 network and other detection networks on this dataset. At the same time, we tested the network's generalization ability on the public MSTAR dataset.

A. Data Collection

We use the P440 radar (Time Domain, Inc.,) [42] to collect the echo signals of multiple people engaging in simultaneous motion under indoor wall penetration conditions to obtain an overall time–distance image. The radar operates at a central frequency of 4.3 GHz, with a frequency band range from 3.1 to 4.8 GHz and a sampling frequency of 16 GHz. The data collection scenario is shown in Fig. 6. The directional transmit and receive antenna is positioned 1 m above the ground, in close proximity to a concrete slab with a thickness of 12 cm. The echoes received by the radar are transmitted to a laptop via Ethernet. The detection scene size is approximately 10×10 m^2, and the target activity area is located 2–9.5 m from the radar. Nine adults participated in the data collection, covering eight common human actions. The action types and descriptions are shown in Table I. Before each acquisition, we randomly selected 2–3 participants. The selected participants randomly chose different positions (radial distances) within the detection area and independently selected one of the eight actions to perform. Then, each subject simultaneously performed their chosen action, and data acquisition commenced. Each data acquisition session lasted 10–15 s. We repeated this process multiple times, resulting in a total of 900 feature maps and ensuring a relatively balanced number of samples for each action type. The dataset was divided into training, validation, and test sets with a ratio of 7:2:1.

B. Data Processing and Annotation

Preprocessing of the collected radar data is essential to enhance signal quality and extract meaningful features for action recognition. We perform signal preprocessing by clutter removal and adaptive amplitude balancing. The preprocessed echo matrix is used to generate the time–distance image, which reflects the target motion characteristics through scattering point distribution and distance changes.



Fig. 6. Data collection scenario.

The raw radar signals include cluttered components from static objects (e.g., walls) and environmental noise, which can mask the dynamic signatures of human activities. We use the moving target indication (MTI) [43] and average subtraction filtering technology [44] to remove static target reflections and random noise. In addition, due to the influence of factors such as distance attenuation, action type, and individual differences, there are large differences in the echo amplitudes of multiple targets. Compensating only by distance results in an excessive gain of far-field noise, while manual adjustment of echo amplitude requires additional a priori information. To solve this problem, we introduce an adaptive amplitude balancing algorithm. This algorithm can automatically adjust the intensity of echo signals, especially to enhance weak signals, thereby improving the model's ability to resolve targets. The specific processing process is shown in Algorithm 1. Finally, we used the Roboflow tool [45] to manually annotate each subject's positioning box and category label in the time–range image. The distance dimension is used as the x-axis, and the time dimension is used as the y-axis.

The data preprocessing and annotation are illustrated in Fig. 7(a), which includes the raw time–range image, the preprocessed image after clutter removal and adaptive amplitude balancing, and the time–range image with annotations. An example of the time–range image for eight action categories is shown in Fig. 7(b). It can be seen that the clutter removal operation effectively eliminates coupling signals and reflections from static targets such as walls. At the same time, adaptive amplitude balancing not only enhances weak signal components but also ensures that noise remains low.

C. Data Augmentation

Limited data and single scenarios will lead to weak model generalization capabilities. To enhance the robustness of the model, we performed data augmentation on the training set. The data-enhanced sample is the time–distance image obtained after the above preprocessing. Methods include adding 5% Gaussian noise to simulate the influence of radar

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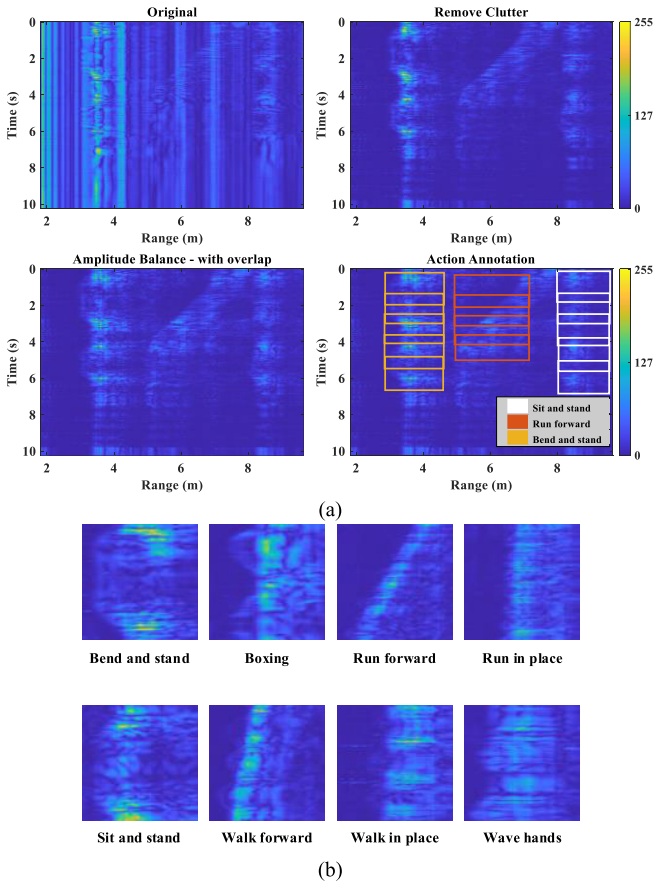


Fig. 7. Schematic of data processing and annotation. (a) Original, pre-processed, and annotated time–range images. (b) Time–range images for eight types of actions.

noise environment, flipping the time–range image vertically to change the time order, using cutout to randomly block several parts of the target feature map to simulate a partial occlusion scene, and randomly scaling and cropping the time–range image to obtain samples from different time and space observation ranges. Finally, we randomly selected four pictures, cropped and scaled them appropriately, and then spliced them into one picture to introduce realistic multi-person action overlay data. After data augmentation, the capacity of the training set is expanded three times, while the samples of the validation set and test set remain unchanged.

D. Experimental Results

We implement the improved YOLOv5 network based on the Pytorch framework and train the network on RTX 3090 GPU. The training strategy is the SGD optimizer, with an initial learning rate of 0.01. In the later stage, the cosine annealing strategy adjusts the learning rate, and the batch size is set to 24. Evaluation indicators include precision (P), recall rate (R), $F1$-score, and mean average precision (mAP).

We tested the performance of the original YOLOv5 and improved YOLOv5 on the test set, as shown in Table II. It can be seen that the improved YOLOv5 network exceeds the original network in both mAP and recall, and the recognition accuracy is significantly improved. It shows that adding the BiFPN feature enhancement module, CA attention mechanism, and SIoU loss function enhances YOLOv5's ability to process complex radar feature images. Fig. 8 illustrates a comparison of the detection and classification results on a specific test sample, showcasing the performance of the original YOLOv5 model versus our improved version. We can see that the improved YOLOv5 gives more accurate bounding box positioning and classification predictions. We also evaluated the detection metrics for each action category involving multiple persons, including mAP, precision, recall, and $F1$-score, as shown in Table III. It can be observed that the detection accuracy for running and walking postures (such as "walking in place," "walking forward," "running in place," and "running forward") is relatively high. In contrast, the accuracy for detecting "boxing" and "waving" actions is noticeably lower. We speculate that this disparity may be due to the nature of the movements involved. Running and walking typically engage the entire body, resulting in more pronounced energy reflections in the echo signals, which lead to clearer texture features in the time–distance images. In contrast, "boxing" and "waving" actions are primarily concentrated in the arm region, occupy a smaller physical area, and generate weaker signal energy. This causes the texture features in the time–distance images less distinct and makes position labeling more difficult, thus making it more challenging for the model to distinguish these actions.

In addition, we performed the following further experiments.

1) Ablation Experiment: BiFPN, CA, and SIoU were gradually added to the baseline model to test the network

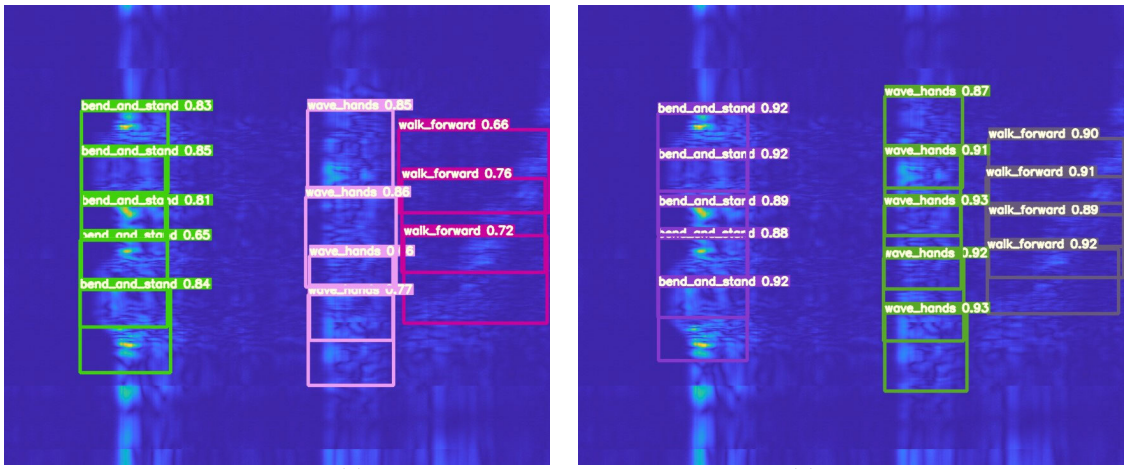
TABLE II
RESULTS OF ABLATION EXPERIMENTS ON VARIOUS COMPONENTS

Model	mAP @0.5	P	R	F1
YOLOv5	0.634	0.59	0.682	0.633
YOLOv5+CA	0.647	0.426	0.758	0.546
YOLOv5+BiFPN+CA	0.659	0.483	0.735	0.583
YOLOv5+BiFPN+CA +SIoU Loss	0.669	0.482	0.758	0.589

TABLE III
METRICS FOR TARGET DETECTION IN VARIOUS ACTIONS

Class	mAP @0.5	P	R	F1
Bend and stand	0.726	0.603	0.824	0.696
Boxing	0.384	0.379	0.413	0.395
Run forward	0.793	0.448	0.901	0.598
Run in place	0.891	0.567	0.923	0.702
Sit and stand	0.639	0.512	0.737	0.604
Walk forward	0.776	0.548	0.861	0.670
Walk in place	0.703	0.439	0.812	0.570
Wave hands	0.436	0.36	0.594	0.448
All	0.669	0.482	0.758	0.589

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(a) (b)

Fig. 8. Examples of detection results of original and improved YOLOv5 models. (a) Baseline YOLOv5 detection results. (b) Improved YOLOv5 detection results.

TABLE IV
MODEL PERFORMANCE COMPARISON

Model	mAP @0.5	P	R	F1
YOLOv3 [46]	0.618	0.602	0.624	0.613
Faster R-CNN [47]	0.566	0.495	0.657	0.564
Cascade R-CNN [48]	0.575	0.560	0.672	0.611
RetinaNet [49]	0.343	0.331	0.389	0.358
SSD [50]	0.558	0.575	0.530	0.552
Ours	0.669	0.482	0.758	0.589

TABLE V
EXPERIMENTAL RESULTS ON THE MSTAR DATASET

Model	mAP @0.5	P	R	F1
YOLOv3	0.705	0.633	0.629	0.631
Faster R-CNN	0.834	0.881	1.000	0.863
YOLOv5	0.861	0.683	0.832	0.750
Ours	0.913	0.754	0.899	0.820

performance, and the results are shown in Table II. It can be seen from the results that CA provides an mAP improvement of about 1.3%, while BiFPN further improves the accuracy by about 1.2%. The SIoU loss increases the mAP value by 1%. The results show that all three have a positive effect on improving detection and classification accuracy. The BiFPN module enhances the expression of multiscale features and improves the network's ability to model human action temporal distance information. The SIoU loss function optimizes bounding box regression and reduces false and missed detections. The CA attention mechanism improves discriminability by suppressing irrelevant features and enhances the model's ability to recognize complex actions.

2) Comparative Experiment: We also compared the improved network with other deep learning-based multi-person action recognition methods, as shown in Table IV. The results show that, in complex environments, the improved YOLOv5 is better than other detection networks, proving that its detection ability for human action time–range images is stronger.

3) Generalization Testing: In addition to time–range feature images, synthetic aperture radar (SAR) images can also be used to detect targets [51]. The public MSTAR dataset was used to test the network generalization ability [52], [53]. This dataset comprises SAR images featuring ten types of military targets captured under various azimuth angles and radio frequency conditions. Fig. 9 presents the optical and SAR images of these military targets. We directly train the improved YOLOv5 network and other comparison models on this dataset, and the results are shown in Table V. Our model outperforms other models. Compared with the baseline, the mAP value of the improved YOLOv5 network increased by 5.2%. This verifies the strong generalization performance of the proposed model. Combined with the detection result example in Fig. 10, the improved model successfully detected multiple targets and gave accurate classifications.

In summary, quantitative indicator comparison, intuitive case analysis, ablation experiments, and cross-dataset testing prove that the proposed improved YOLOv5 network can achieve good action detection and classification performance in radar penetration and multi-person complex environments. The addition of each module can improve network expression capabilities.

V. LIMITATIONS AND FUTURE WORK

Despite the good results obtained, the current work has some limitations.

1) *Dataset Scope:* The action categories we collected are relatively limited, including only eight categories of

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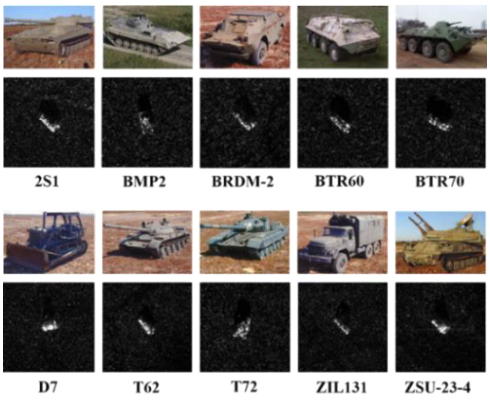


Fig. 9. Examples of optical and SAR images of military equipment from the MSTAR dataset.

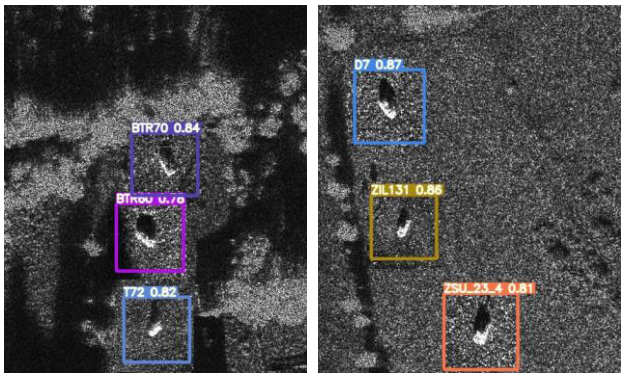


Fig. 10. MSTAR dataset detection result.

daily actions. The size of the dataset is also limited. Building databases on a larger scale and with more categories will further validate the algorithm's performance.

2) *Fusion of Multiple Radar Data:* Multiple radars can provide information from multiple perspectives, improve recognition accuracy, and especially alleviate the problem of accuracy decline caused by occlusion.
3) *Adaptability of the Model:* Although our model performs well on the current dataset, its adaptability in different environments has not been fully verified. To enhance the model's generalization ability, we plan to collect data from multiple environments (such as outdoors and different walls) to deal with more practical application scenarios.
4) *Deployment of the Model on Embedded Devices:* In the current work, we do not delve into the actual deployment of the model on embedded devices, which may affect its usefulness in some real-world scenarios.

This research opens a new direction for multi-person action detection and recognition in complex environments sensed by IR-UWB radar. As hardware conditions improve and datasets expand, a large number of new problems and challenges can still be explored in this field. We will work hard to improve the current shortcomings and look forward to further promoting technological progress in this field.

VI. Ethical Considerations

It is important to note that while YOLO is typically associated with optical image detection, in this work, we apply it to radar time–range images. These images do not contain personal visual information, thereby mitigating privacy concerns associated with image-based detection. Nonetheless, they still represent movement patterns that could be sensitive, so it is necessary to ensure that all research activities comply with ethical standards and regulations. In our study, data collection was conducted with the informed consent of the participants, and the data were used solely for research purposes. This research has been approved by the Nanjing University of Science and Technology Ethics Committee.

VII. Conclusion

Given the limitation that conventional IR-UWB radar systems are difficult to adapt to complex environments, this article proposes an improved YOLOv5 network to achieve multi-person action detection and classification under wall penetration conditions. The main contributions include: 1) constructing a multi-person action dataset based on IR-UWB radar under through-wall conditions; 2) introducing CA attention, BiFPN module, and SIoU loss based on YOLOv5 to improve the feature expression ability of the model; and 3) the experimental results prove the effectiveness of the proposed method. The improved YOLOv5 model shows better performance and generalization capabilities in multitarget complex scenes than other detection networks. This study initially explores a powerful action understanding model based on deep learning in the complex environment of radar perception and also lays the foundation for subsequent mining of richer action semantics.

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